

GIA MINH HOANG, Ph.D.

Postdoctoral Fellow
College of Medicine and Science
Mayo Clinic Arizona

Phone : 480-527-3052
E-mail : hoang.giaminh@mayo.edu

Google Scholar: <https://scholar.google.com/citations?user=Lo1YsfcAAAAJ>

Website: <https://giaminhgist.github.io/>

RESEARCH INTERESTS

My research interests revolve around developing innovative AI tools and promoting their generalizability and trustworthiness on biomedical imaging data. Research topics include, but are not limited to:

- Robust and trustworthy AI application for **medical imaging AI** (MRI/CT/PET): generalization under **domain shift**, external validation, and bias/uncertainty quantification
- **Explainable AI** for imaging biomarkers and clinically meaningful model interpretation
- **Digital twins** for patient-/organ-specific modeling: physiology-informed, data-driven twins for organ perfusion and therapy planning
- Multimodal imaging + physiological data integration for diagnosis, prognosis, and treatment response modeling

EDUCATION AND TRAINING

Postdoctoral Researcher

Aug. 2025 – Present

Department of Physiology and Biomedical Engineering
Mayo Clinic, Arizona

Gwangju Institute of Science and Technology

Feb. 2022 – Aug. 2025

Ph.D. in Biomedical Engineering

Advisor: Jae Gwan Kim, Ph.D.

Dissertation: A Deep Learning-Based Approach for Early Alzheimer's Disease Diagnosis: Addressing Generalization and Domain Shift Across Populations

Gwangju Institute of Science and Technology

Aug. 2019 – Aug. 2021

M.S. in Biomedical Engineering

Hanoi University of Science and Technology

Aug. 2014 – Aug. 2019

B.S. in Biomedical Engineering

TEACHING EXPERIENCE

Gwangju Institute of Science and Technology

Graduate Teaching Assistant

Graphical System Design Programing (Fall 2023)

AI Experience Lab (Fall 2024)

PROJECT EXPERIENCES

Development of artificial intelligence approaches for diagnosis progression of Mild cognitive impairment to Alzheimer's disease using medical imaging

Jan. 2021 – Aug. 2025

Gwangju Institute of Science and Technology

Role: Research Assistant

Development of a new electronic drug for Alzheimer's disease using transcranial ultrasound and optical stimulation

Feb. 2020– Aug. 2025

Gwangju Institute of Science and Technology

Role: Research Assistant

ReLiverate: Digital twin model for regenerating Discarded Human Livers

Aug. 2025 - Present

Mayo Clinic Scottsdale Campus

Role: Postdoctoral Researcher

TECHNICAL SKILLS

- Hands-on experience with medical image analysis on MRI.
- Proficiency in Python and common ML/DL frameworks (PyTorch, TensorFlow)
- Hands-on experience with neuroimaging processing pipelines (FreeSurfer, FSL)
- Excellent writing and communication skills, especially research manuscript/publications.

PUBLICATIONS IN SCI(E) JOURNALS

1. Tam Dao*, Thanh T. Nguyen*, **Gia Minh Hoang***, Junhyeon Park, Yunju Jo, Thach Hoang Ngoc, Diep Hong Pho, Dien Tran Minh, Emma Anh Ton, Sunjae Lee, Hyun Jin Kim, Vu Chi Dung, Jae Gwan Kim, Dongryeol Ryu. "*Nuclear enlargement as a histological hallmark of skeletal muscle aging, revealed by deep learning-driven analysis and validated in inflammatory myopathies.*" **Aging Cell**, June 2026.
2. **Gia Minh Hoang** and Jae Gwan Kim. "*High-generalizability Deep Learning Framework for Early Detection of Mild Cognitive Impairment Progression to Alzheimer's Disease.*" **Alzheimer's & Dementia** Dec 2025. https://doi.org/10.1002/alz70856_101596
3. **Gia Minh Hoang** and Jae Gwan Kim. "*Explainable Deep Learning for Multi-Cohort Alzheimer's Disease Classification using MRI: Insights into Pathological Brain Regions.*" **Alzheimer's & Dementia** Dec 2025. https://doi.org/10.1002/alz70856_101859
4. Le Thien Phuc Nguyen*, Thien Nguyen*, Thanh-Huy Nguyen, **Gia Minh Hoang**, Anh Mai Vu, Ulas Bagci. "*QwenVLConnector: A Fast, Unified Medical VLM Chatbot for Fine-Grained Clinical Perception and Text Generation.*" **MICCAI 2025**, FLARE Challenge. Sep 2025. <https://openreview.net/forum?id=FZZyXegzeQ>
5. **Gia Minh Hoang**, Nhat Hoang, Tran Thi Hoai Thu, Hoang Tien Trong Nghia and Jae Gwan Kim. "*AttCORAL: Domain-Adaptive Attention Networks for Early Alzheimer's*

- Disease Diagnosis*". **IEEE Access**, June 2025. <https://doi.org/10.1109/ACCESS.2025.3580780>
6. **Gia Minh Hoang**, Youngjoo Lee, Jae Gwan Kim. "Deep learning with guided attention for early diagnosis of Alzheimer's disease." **Physica Scripta**, Apr 2025. <https://doi.org/10.1088/1402-4896/add2a6>
 7. **Gia Minh Hoang**, Youngjoo Lee, Jae Gwan Kim. "A reproducible 3D convolutional neural network with dual attention module (3D-DAM) for Alzheimer's disease classification." **arXiv**, Oct 2023. <https://doi.org/10.48550/arXiv.2310.12574>
 8. **Gia Minh Hoang**, Ue-Hwan Kim, Jae Gwan Kim. "Vision transformers for the prediction of mild cognitive impairment to Alzheimer's disease progression using mid-sagittal sMRI." **Frontiers in Aging Neuroscience**, Apr 2023. <https://doi.org/10.3389/fnagi.2023.1102869>
 9. Mincheol Park*, **Gia Minh Hoang***, Thien Nguyen, Eunkyung Lee, Hyun Jin Jung, Youngshik Choe, Moon Hwan Lee, Jae Youn Hwang, Jae Gwan Kim, Tae Kim. "Effects of transcranial ultrasound stimulation pulsed at 40 Hz on A β plaques and brain rhythms in 5 $\times$ FAD mice." **Translational Neurodegeneration**, Dec 2021. <https://doi.org/10.1186/s40035-021-00274-x>

PROFESSIONAL SERVICE

Peer Reviewer

- *Computers in Biology and Medicine*
- *IEEE Journal of Biomedical and Health Informatics*
- *Engineering Research Express*
- *Physics in Medicine & Biology*
- *Machine Learning: Science and Technology*
- *Physica Scripta*

HONOR & AWARDS

- Excellent Students Award, Hanoi University of Science and Technology (2018)
- Scholarship for Excellent Students, Hanoi University of Science and Technology (2019)
- Best Researcher Award, GIST (2022)
- Lotte International Student Scholarship Award, Lotte Scholarship Foundation (2024)
 - **~35,000\$ for 4 years**
- Best Researcher Award, GIST (2025)

REFERENCES

Jae Gwan Kim, Ph.D.

Address: 123 Cheomdan-gwagi-ro, Buk-gu Gwangju, South Korea 61005, Department of Biomedical Science & Engineering, Gwangju Institute of Science and Technology

Email: jaekim@gist.ac.kr

Tuan Anh Le, Ph.D.

Address: 13400 E Shea Blvd 85259 Scottsdale, AZ, Department of Physiology and Biomedical Engineering, Mayo Clinic College of Medicine and Science
Email: Le.TuanAnh@mayo.edu

Hakan Ceylan, Ph.D.

Address: 13400 E Shea Blvd 85259 Scottsdale, AZ, Department of Physiology and Biomedical Engineering, Mayo Clinic College of Medicine and Science
Email: Ceylan.Hakan@mayo.edu